Hi. I’m Adriel, and in this text, I’m going to tell you about the implications and problems I’ve found solving this problem.

Before anything, I read the texts files on the mail, I started watching the datasets on the csv files. First thing I’ve noticed was the mixture of numerical, categorical and coordinates separated into various fields, so an encoding method was going to be needed.

The encoding method.

pickup	delivery	equipment
Target encoding	Target encoding	One-hot encoding

All of the encoding was made in excel, totally a comfort pick. The target encoding was made using conditional means and XLOOKUP into a dictionary sheet that can be found in train_corrected.xlsx

The one-hot encoding was made in the same dictionary page and linked using INDEX/MATCH functions, the dictionary took the next form

Dry Van	1	0	0
Flatbed	0	1	0
Reefer	0	0	1

The split(s).

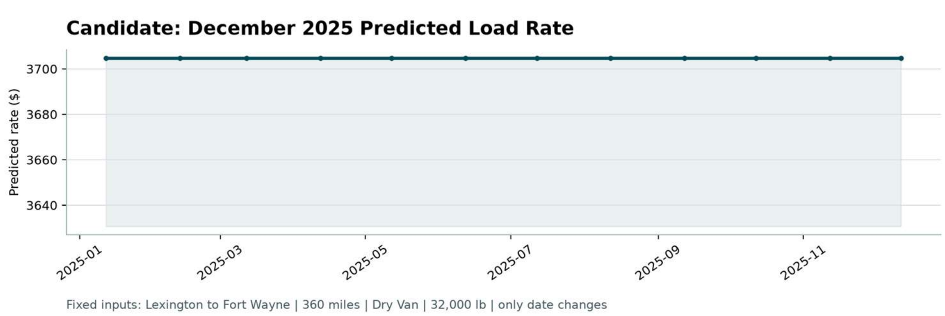
For all the missing values, I used k-neighbors imputation using k=6 after testing k= 3,4,5 to, the best results in term of LOSS/MSE were achieved by the k=6 imputations.

I tried three models (mentioned below) to finally choose a neural network model using sklearn and keras (the one on the solution code), data was split into 80 train and 20 test, Pareto's law mentioned.

The linear model had similar results to the neural network one, the difference and breaking point was that there was no homoscedasticity and the residuals were not normally distributed.

Also, I tried LASSO regression but the penalization λ was really close to 0 in every k imputations, the highest was 0.31.

The december score.



More on this in the code…